

Anonymous ARR Rebuttal

Anonymous

1 Additional materials

Model	COH	SYN	PHR	VOC	GRA	CON	Avg.
ModernBERT (Ours)	0.65	0.68	0.66	0.68	0.68	0.70	0.67
NLA (Ours)	0.59	0.65	0.49	0.64	0.47	0.64	0.58
RAvg (Ours)	0.62	0.65	0.61	0.62	0.68	0.64	0.64
Q-CFA (Ours)	0.62	0.64	0.63	0.61	0.69	0.65	0.64
Q-MIRT (Ours)	0.62	0.64	0.65	0.62	0.68	0.64	0.64
BERT [1]	0.57	0.66	0.65	0.63	0.63	0.66	0.63
RoBERTa [1]	0.62	0.67	0.69	0.65	0.71	0.71	0.67
MTL _{vanilla} [1]	0.63	0.69	0.72	0.67	0.72	0.71	0.69
MTAA [1]	0.63	0.69	0.71	0.67	0.73	0.72	0.69

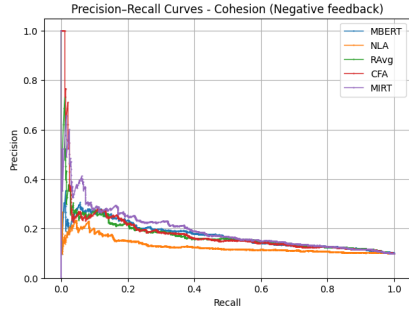
Table 1: Comparison of our ModernBERT-based grader against [1]’s graders in terms of Spearman’s rank coefficient against the original ELLIPSE test set labels.

Model	Analytic Dimension				
	CON	ORG	LAN	VOC	MCH
NLA	21.46	16.34	14.23	20.87	22.66
Q-RAvg	22.01	27.51	15.37	16.31	25.87
Q-WAvg	23.36	29.77	14.90	14.86	29.47
Q-CFA	21.94	24.27	15.35	14.52	17.84
Q-MIRT	20.90	28.44	16.11	18.20	19.12

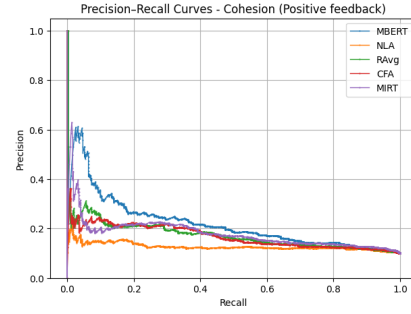
Table 2: Self-referential evaluation in terms of $F_{0.5}$ for **Negative** feedback (IC-NALE EE) using **Qwen 2.5 72B**.

Model	Analytic Dimension				
	CON	ORG	LAN	VOC	MCH
NLA	14.15	27.23	24.26	12.32	18.82
Q-RAvg	17.36	20.41	26.59	22.46	19.94
Q-WAvg	16.31	21.22	26.16	19.21	19.07
Q-CFA	14.17	20.64	24.75	18.03	18.67
Q-MIRT	19.93	19.43	23.52	19.86	14.86

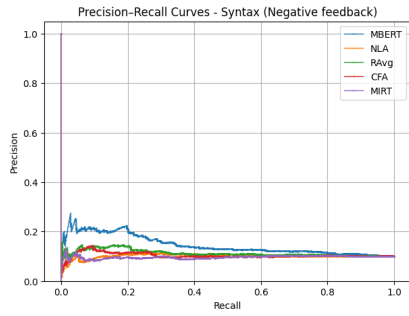
Table 3: Self-referential evaluation in terms of $F_{0.5}$ for **Positive** feedback (IC-NALE EE) using **Qwen 2.5 72B**.



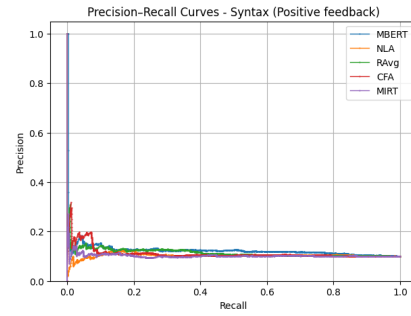
(a) Cohesion - negative feedback



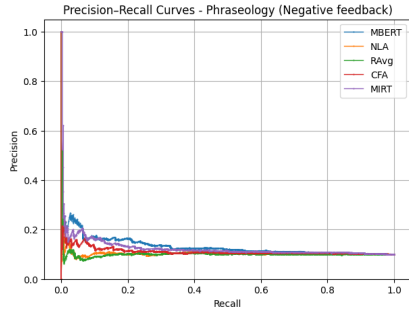
(b) Cohesion - positive feedback



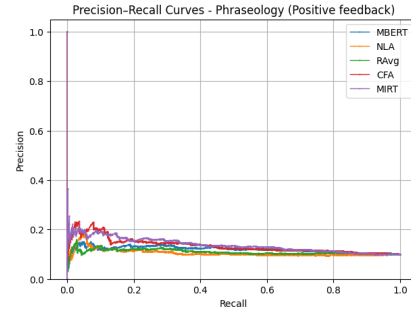
(c) Syntax - negative feedback



(d) Syntax - positive feedback

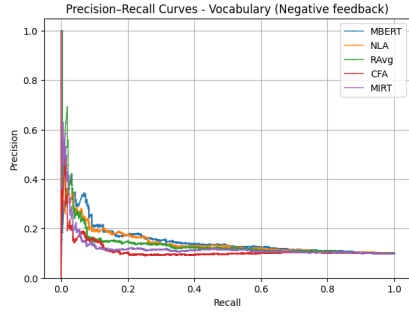


(e) Phraseology - negative feedback

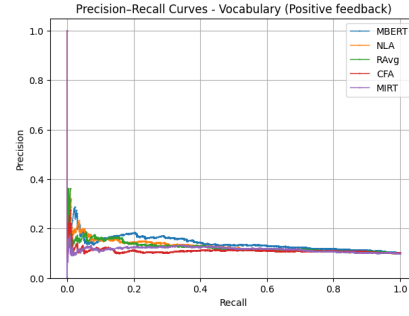


(f) Phraseology - positive feedback

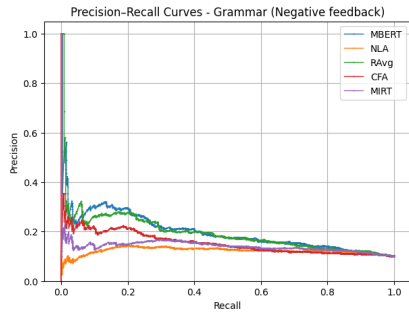
Figure 1: Precision-recall curves for self-referential evaluation results on EL-LIPSE using GPT-4.1-mini (1 of 2).



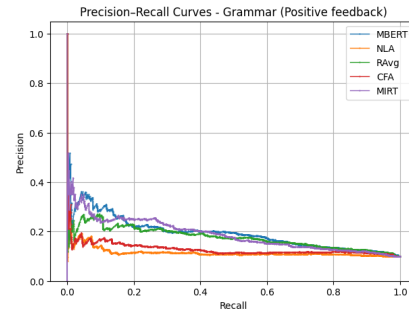
(a) Vocabulary - negative feedback



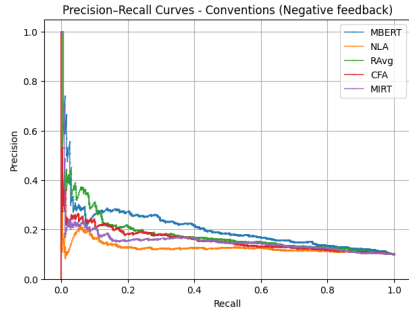
(b) Vocabulary - positive feedback



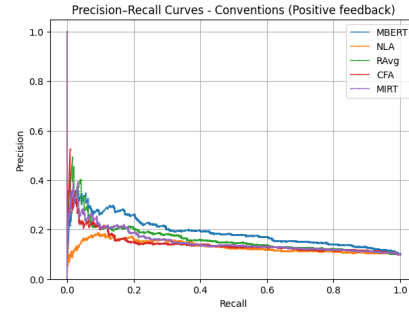
(c) Grammar - negative feedback



(d) Grammar - positive feedback



(e) Conventions - negative feedback



(f) Conventions - positive feedback

Figure 2: Precision-recall curves for self-referential evaluation results on EL-LIPSE using GPT-4.1-mini (2 of 2).

References

- [1] S. Chen, Y. Lan, and Z. Yuan, “A multi-task automated assessment system for essay scoring,” in *International Conference on Artificial Intelligence in Education*, pp. 276–283, Springer, 2024.